

Zhiyuan Li

Postdoctoral Researcher

Department of Electrical
Engineering and Automation
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Research Interests

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Multi-Agent Systems, Reinforcement Learning, Robotics, Foundation Models

Education

University of Electronic Science and Technology of China <i>Ph. D in Computer Science and Technology</i>	Sep 2019 – July 2024 Chengdu China
North China Electric Power University <i>Bachelor in Computer Science and Technology</i>	Sep 2015 – July 2019 Beijing China

Research Presentations

Online Coordinated NFV Resource Allocation via Novel Machine Learning Techniques

Zhiyuan Li, Lijun Wu, Xiangyun Zeng, Xiaofeng Yue, Yulin Jing, Wei Wu, and Kaile Su.
[TNSM'22] *IEEE Transactions on Network and Service Management*, 2022.

Backpropagation Through Agents

Zhiyuan Li, Wenshuai Zhao, Lijun Wu, and Joni Pajarinen.
[AAAI'24] *Thirty-Eighth AAAI Conference on Artificial Intelligence*, 2024.

Coordination as Inference in Multi-Agent Reinforcement Learning

Zhiyuan Li, Lijun Wu, Kaile Su, Wei Wu, Yulin Jing, Tong Wu, Weiwei Duan, Xiaofeng Yue, Xiyi Tong, and Yizhou Han.
[Neural Networks'24] *Neural Networks*, 2024.

Optimistic Multi-Agent Policy Gradient for Cooperative Tasks

Wenshuai Zhao, Yi Zhao, Zhiyuan Li, Juho Kannala, and Joni Pajarinen.
[ICML'24] *The Forty-first International Conference on Machine Learning*, 2024.

AgentMixer: Multi-Agent Correlated Policy Factorization

Zhiyuan Li, Wenshuai Zhao, Lijun Wu, and Joni Pajarinen.
[AAAI'25] *Thirty-Ninth AAAI Conference on Artificial Intelligence*, 2025.

Learning Progress Driven Multi-Agent Curriculum

Wenshuai Zhao, Zhiyuan Li, and Joni Pajarinen.
[ICML'25] *The Forty-Second International Conference on Machine Learning*, 2025.

Rethinking Temporal Consistency in Video Object-Centric Learning: From Prediction to Correspondence

Zhiyuan Li, Rongzhen Zhao, Wenyan Yang, Wenshuai Zhao, Pekka Marttinen, and Joni Pajarinen.
[ICML'26] *The Forty-Third International Conference on Machine Learning*, 2026.

Multi-agent neighborhood coordinated and holistic optimized actor-critic framework for adaptive traffic signal control

Qi Deng, Lijun Wu, Zhiyuan Li, Kaile Su, Wei Wu, and Weiwei Duan.

[Applied Intelligence] The International Journal of Research on Intelligent Systems for Real Life Complex Problems, 2025

Closed-Loop Vision-Language Planning for Multi-Agent Coordination

Zhiyuan Li, Wenshuai Zhao, and Joni Pajarinen.

[SE@AAMAS'26] The 25th International Conference on Autonomous Agents and Multiagent Systems, 2026.

Bridging the Embodiment Gap: Disentangled Cross-Embodiment Video Editing

Zhiyuan Li, Wenyan Yang, Wenshuai Zhao, Yue Ma, Yuanpeng Tu, Pekka Marttinen, and Joni Pajarinen.

[NeurIPS'26] The Fortieth Annual Conference on Neural Information Processing Systems, 2026. (Under Review)

Why Cross-Skeleton Retargeting Is Non-Identifiable: Structural Limits of Generative Motion Models

Zhiyuan Li, Wenyan Yang, Pekka Marttinen, and Joni Pajarinen.

[NeurIPS'26] The Fortieth Annual Conference on Neural Information Processing Systems, 2026. (Under Review)

Do We Really Need the Default Recipe for Video Object-Centric Learning?

Zhiyuan Li, Rongzhen Zhao, Wenyan Yang, Wenshuai Zhao, Arno Solin, Juho Kannala, Pekka Marttinen, and Joni Pajarinen.

[TPAMI] IEEE Transactions on Pattern Analysis and Machine Intelligence. (Under Review)

Multiple robots path planning based on improved artificial potential field and multi-agent reinforcement learning

Qingfeng Yao, Qifeng Zhang, Qiang Li, Zhiyuan Li, Linghan Meng, Yingzhe Sun, and Cong Wang.

[TCDS] IEEE Transactions on Cognitive and Developmental Systems. (Under Review)

Awards & Honors

Scholarship under China Scholarship Council

Ministry of Education of the P. R. China

2022

Academic Scholarship

University of Electronic Science and Technology of China

2019-2022

Honorable Mention of Mathematical Contest in Modeling

2017